Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is i^2? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **By definition $i^2 = -1$. The student dropped the negative sign.**

Correct answer: -1

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify $\sqrt{(-9)}$. Student's answer: ± 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$\sqrt{(-9)} = \sqrt{9} \cdot \sqrt{(-1)} = 3i$. Real square roots of negatives do not exist.**

Correct answer: **$3i$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is i^3? Student's answer: i

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$i^3 = i^2 \cdot i = -1 \cdot i = -i$. The student forgot to apply $i^2 = -1$.**

Correct answer: **$-i$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add (2 + 3i) + (4 - i). Student's answer: 6 + 4i

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The imaginary parts are 3i and −i, which add to 2i. The answer is 6 + 2i.**

Correct answer: **6 + 2i**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(3 + i)(3 - i)$. Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$(3 + i)(3 - i) = 9 - i^2 = 9 + 1 = 10$. The student forgot that $i^2 = -1$.**

Correct answer: **10**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(1 + 2i)(3 + i)$. Student's answer: $5 + 7i$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$2i \cdot i = 2i^2 = -2$. The answer is $3 + 7i - 2 = 1 + 7i$.**

Correct answer: **$1 + 7i$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 + 4 = 0$. Student's answer: $x = \pm 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x^2 = -4$, so $x = \pm 2i$. The student dropped the imaginary unit.**

Correct answer: **$x = \pm 2i$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 + 2x + 5 = 0$. Student's answer: $x = -1 \pm 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The discriminant is 4 − 20 = −16, so the roots are $(-2 \pm 4i)/2 = -1 \pm 2i$.**

Correct answer: **$x = -1 \pm 2i$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 - 6x + 10 = 0$. Student's answer: $x = 3 \pm 2i$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The discriminant is 36 − 40 = −4, so $\sqrt{(-4)} = 2i$. Then $x = (6 \pm 2i)/2 = 3 \pm i$.**

Correct answer: **$x = 3 \pm i$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Rewrite $x^2 + 6x + 1$ by completing the square. Student writes: $(x + 3)^2 - 9$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$(x + 3)^2 = x^2 + 6x + 9$, so $x^2 + 6x + 1 = (x + 3)^2 - 8$. The student forgot to adjust the +1.**

Correct answer: **$(x + 3)^2 - 8$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the maximum value of $f(x) = -2x^2 + 8x$? Student's answer: It has a minimum of 8.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The coefficient -2 is negative, so the parabola opens down and has a maximum. The vertex is at x = 2 with f(2) = 8.**

Correct answer: **Maximum of 8**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the vertex of $y = x^2 - 4x + 7$? Student's answer: $(-2, 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x = -(-4)/2 = 2$. Then $y = 4 - 8 + 7 = 3$. The vertex is $(2, 3)$.**

Correct answer: **$(2, 3)$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $p(x) = x^2 + 3x - 4$. What is $p(1)$? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$p(1) = 1 + 3 - 4 = 0$. The student did not finish the arithmetic.**

Correct answer: 0

Find the Error

Grade 11 · A.APR.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the remainder when $x^3 - 2x + 1$ is divided by $x - 2$? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$p(2) = 8 - 4 + 1 = 5$. The remainder is 5.**

Correct answer: **5**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $x - 3$ a factor of $x^2 - 5x + 6$? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$p(3) = 9 - 15 + 6 = 0$, so $x - 3$ is a factor.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $6/x = 3$. Student's answer: $x = 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: 6 = 3x, so x = 2. The student divided 6 by 2 incorrectly.

Correct answer: x = 2

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x/(x - 2) = 2$. Student's answer: $x = 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x = 2(x - 2)$ gives $x = 4$. The value $x = 2$ makes the denominator zero, so it is not allowed.**

Correct answer: **$x = 4$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $1/x + \frac{1}{2} = 1$. Student's answer: $x = 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$1/x = \frac{1}{2}$ gives $x = 2$. The student combined the fractions incorrectly.**

Correct answer: **$x = 2$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = x^2$ and $y = x + 2$. Student's answer: (2, 4) only

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x^2 - x - 2 = 0$ gives $x = 2$ or $x = -1$. The second point is (-1, 1).**

Correct answer: **(2, 4) and (-1, 1)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = x^2$ and $y = 4$. Student's answer: (2, 4)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x^2 = 4$ gives $x = 2$ or $x = -2$. Both points (2, 4) and (−2, 4) work.**

Correct answer: **(2, 4) and (−2, 4)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = x^2 - 1$ and $y = 3$. Student's answer: (2, 3)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x^2 = 4$, so $x = \pm 2$. The solutions are (2, 3) and (−2, 3). The student left out one point.**

Correct answer: **(2, 3) and (−2, 3)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the x-values where $y = x^2$ and $y = 2x + 3$ intersect. Student's answer: $x = 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x^2 - 2x - 3 = 0$ gives $x = 3$ or $x = -1$. The student missed $x = -1$.**

Correct answer: **$x = 3$ or $x = -1$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find where y = 2^x and y = x + 3 intersect. Student says: They intersect at x = 3.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **At x = 3, $2^3 = 8$ but 3 + 3 = 6. The graphs cross between x = 2 and x = 3, not at x = 3.**

Correct answer: **Between x = 2 and x = 3**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the x-values where $y = x^3$ and $y = x$ intersect. Student's answer: $x = 1$ only

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$x^3 - x = 0$ gives $x(x - 1)(x + 1) = 0$, so $x = 0, 1,$ or -1.**

Correct answer: **$x = -1, 0, 1$**

Find the Error

Grade 11 · F.IF.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the average rate of change of $f(x) = x^2$ on $[1, 3]$. Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **f(3) − f(1) = 9 − 1 = 8. Divide by 3 − 1 = 2 to get 4.**

Correct answer: **4**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the average rate of change of $f(x) = 3x - 2$ on $[0, 4]$. Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided the change in f by 1 instead of 4. $f(4) - f(0) = 12$, and $12 \div 4 = 3$.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the average rate of change of $f(x) = 2^x$ on $[0, 2]$. Student's answer: 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not divide by 2. $f(2) - f(0) = 4 - 1 = 3$, and $3 \div 2 = 1.5$.**

Correct answer: **1.5**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Where is the vertex of $f(x) = |x - 2| + 1$? Student's answer: $(-2, 1)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$|x - 2| = 0$ when $x = 2$, so the vertex is $(2, 1)$. The student changed the sign of 2.**

Correct answer: **$(2, 1)$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the domain of $f(x) = \sqrt{(x + 4)}$? Student's answer: $x \geq 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: x + 4 ≥ 0 gives x ≥ -4. The student changed the sign.

Correct answer: x ≥ -4

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $f(x) = x + 1$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. What is $f(-2)$? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Since $-2 < 0$, use the first piece. $f(-2) = -2 + 1 = -1$.**

Correct answer: **-1**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 8^x equivalent to 2^(3x)? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $8 = 2^3$, so $8^x = (2^3)^x = 2^{(3x)}$. The two expressions are equivalent.

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A population is modeled by y = 1000(1.05)^t. What is the annual growth rate? Student's answer: 1.05%

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The base is 1.05, so the growth rate is 0.05, or 5% per year. The student used the base as a percent.**

Correct answer: **5%**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Rewrite 3^(2t) in the form b^t. Student's answer: 6^t

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **3^(2t) = (3²)^t = 9^t. The student multiplied the bases.**

Correct answer: **9^t**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$f(x) = 2x + 5$ and g has a constant rate of change of 3. Which function has the greater rate of change? Student's answer: f

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **f has rate 2 and g has rate 3. Function g has the greater rate.**

Correct answer: **g**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which grows faster eventually: $f(x) = 2^x$ or $g(x) = x^2$? Student's answer: $g(x) = x^2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **At x = 10, $2^{10} = 1024$ while $10^2 = 100$. Exponential functions eventually grow faster.**

Correct answer: **$f(x) = 2^x$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$y = 4x + 1$ and $y = x^2 + 3$ are compared. Which has the greater y-intercept? Student's answer: $y = 4x + 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The linear function has y-intercept 1. The quadratic has y-value 0 + 3 = 3 when x = 0, so $y = x^2 + 3$ has the greater intercept.**

Correct answer: **$y = x^2 + 3$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $f(x) = x^2$ and $g(x) = f(x) + 3$. What is $g(2)$? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 3. $f(2) = 4$, so $g(2) = 4 + 3 = 7$. The graph shifts up 3.**

Correct answer: 7

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $g(x) = f(x - 2)$ with $f(x) = x^2$. What is $g(3)$? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not subtract 2. $g(3) = f(1) = 1$.**

Correct answer: 1

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $g(x) = 2f(x)$ with $f(x) = x^2$. What is $g(3)$? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not multiply by 2. f(3) = 9, so g(3) = 18.**

Correct answer: 18

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the inverse of $f(x) = 2x + 3$. Student writes: $f^{-1}(x) = (x + 3)/2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$y = 2x + 3$ gives $x = 2y + 3$, so $y = (x - 3)/2$. The student added 3 instead of subtracting.**

Correct answer: **$f^{-1}(x) = (x - 3)/2$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $f(x) = x^3$, what is $f^{-1}(8)$? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student squared 8 instead of taking a cube root. Since $2^3 = 8$, $f^{-1}(8) = 2$.**

Correct answer: 2

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $f(x) = x^2$ one-to-one over all real numbers? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **f(2) = f(-2) = 4. Two inputs give one output, so f is not one-to-one and has no inverse on all reals.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is $\log_2 8$? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$2^3 = 8$, so $\log_2 8 = 3$. The student used $2^2 \times 2$.**

Correct answer: **3**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is $\log_{10} 1000$? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $10^3 = 1000$, so the answer is 3. The student counted only two zeros.

Correct answer: 3

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $2^x = 32$. Student's answer: $x = 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$2^5 = 32$, so $x = 5$. The student used $2^4 = 16$ as the target.**

Correct answer: **$x = 5$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Convert 180° to radians. Student's answer: 2π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **180° equals π radians. The student used the value for a full circle, which is 360°.**

Correct answer: **π**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Convert 90° to radians. Student's answer: π/3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: $\frac{90}{180} = \frac{1}{2}$**, so 90° = π/2 radians.**

Correct answer: π/2

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An arc has radius 5 and central angle 2 radians. What is the arc length? Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added r and θ. The arc length is 5 × 2 = 10.**

Correct answer: **10**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is $\cos(\pi/2)$? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **At $\pi/2$ (90°), the point on the unit circle is (0, 1). The cosine is the x-coordinate, which is 0.**

Correct answer: 0

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is sin π? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **At π (180°), the point is (-1, 0). The sine is the y-coordinate, which is 0.**

Correct answer: 0

Find the Error

Grade 11 · F.TF.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the point on the unit circle at 60°? Student's answer: $(\sqrt{\frac{3}{2}}, \frac{1}{2})$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: At 60°, cos = $\frac{1}{2}$ and sin = $\sqrt{\frac{3}{2}}$. The point is $(\frac{1}{2}, \sqrt{\frac{3}{2}})$.

Correct answer: $(\frac{1}{2}, \sqrt{\frac{3}{2}})$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the amplitude of $y = 3\sin(2x)$? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The amplitude is |3| = 3. The 2 is inside the sine and affects the period.**

Correct answer: 3

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the period of $y = \sin(2x)$? Student's answer: 2π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The period is $2\pi/2 = \pi$. The student did not divide by B.**

Correct answer: π

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the period of $y = \cos(x/3)$? Student's answer: 3π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Here B = $\frac{1}{3}$, so the period is $2\pi \div (\frac{1}{3}) = 6\pi$. The student divided by 3 instead of by $\frac{1}{3}$.**

Correct answer: **6π**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A model is $y = 2x + 1$. What is the predicted y-value when $x = 5$? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: 2(5) + 1 = 11. The student forgot to add 1.

Correct answer: 11

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An actual value is 12 and the predicted value is 11. What is the residual? Student's answer: -1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: 12 − 11 = 1. The student reversed the subtraction.

Correct answer: 1

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A scatter plot has a correlation coefficient of r = 0.95. Is the linear relationship weak or strong?

Student's answer: weak positive

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **r is close to 1, which indicates a strong positive linear relationship.**

Correct answer: **Strong positive**

Find the Error

Grade 11 · S.ID.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A taxi cost is C = 25 + 4m, where m is minutes. What does the slope 4 mean? Student's answer: $25 per minute

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The slope is the rate of change, which is $4 per minute. The 25 is the starting fee.**

Correct answer: **$4 per minute**

Find the Error

Grade 11 · S.ID.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In the same taxi model C = 25 + 4m, what does the intercept 25 mean? Student's answer: the cost per minute

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **When m = 0, the cost is $25, which is the fixed starting fee. The cost per minute is the slope.**

Correct answer: **The starting fee**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A temperature model is T = 80 - 2h, where h is hours. What does the slope mean? Student's answer: The temperature rises 2 degrees per hour.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The slope is -2, so the temperature falls 2 degrees per hour.**

Correct answer: **The temperature falls 2 degrees per hour**